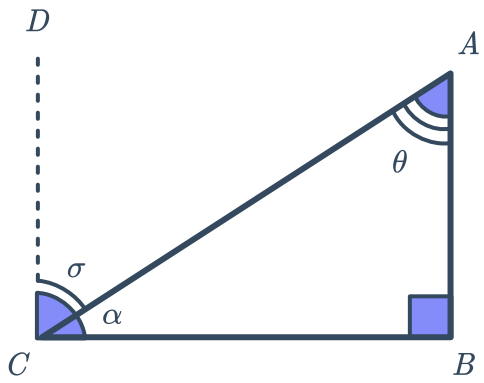
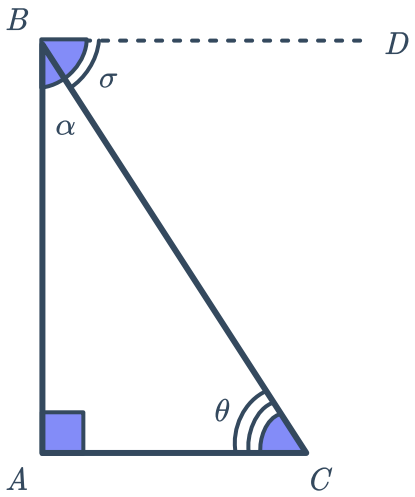
Angles of elevation and depression

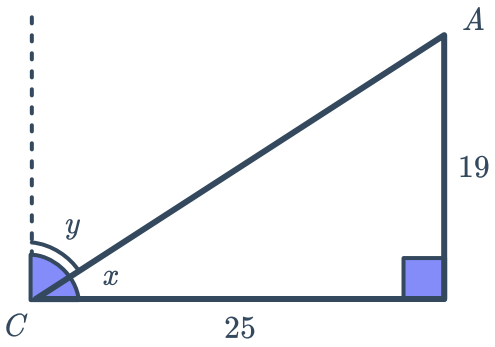
1 A man standing at point C, is looking at the top of a tree at point A. Identify the angle of elevation in the following figure:



2 A man standing at the top of the tower at point B, is looking at the ground at point C. Identify the angle of depression in the following figure:

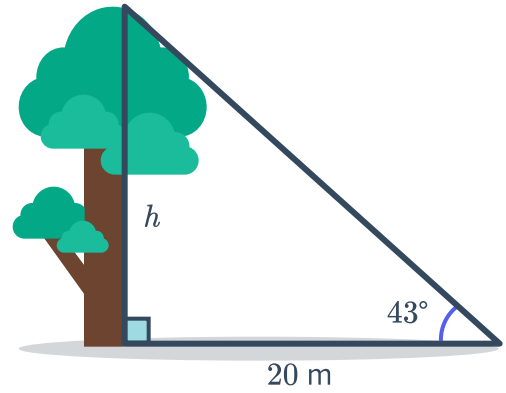


3 Find x, the angle of elevation to point A from point C. Round your answer to two decimal places.

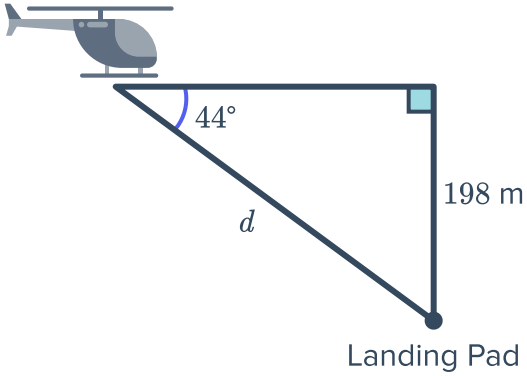


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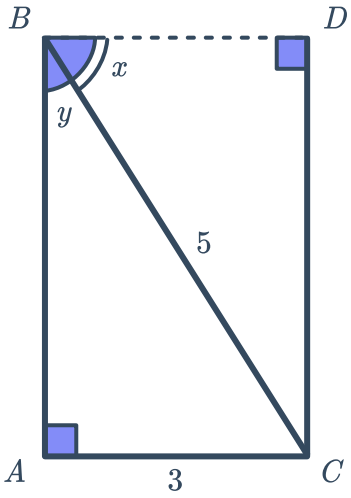
4 Sally measures the angle of elevation to the top of a tree from a point $20\ \text{m}$ away to be 43°.
Find the height of the tree, h, to the nearest whole number.



5 A helicopter is flying at an altitude of $198\ \text{m}$. Its landing pad is at an angle of depression of 44°.
Determine the distance, d, between the helicopter and the landing pad. Round your answer to the nearest whole number.



6 Find x, the angle of depression from point B to point C in the given diagram:
Round your answer to two decimal places.



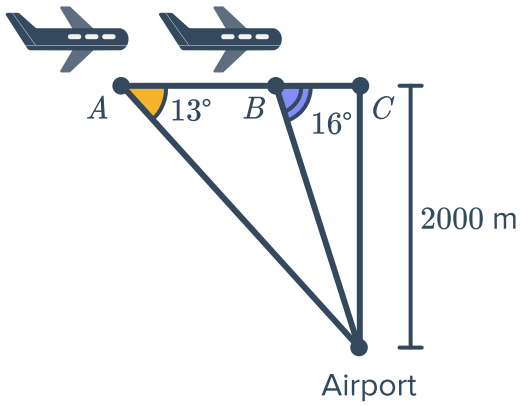
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7 A fighter jet, flying at an altitude of $2000\ \text{m}$ is approaching an airport. The pilot measures the angle of depression to the airport to be 13°. One minute later, the pilot measures the angle of depression again and finds it to be 16°.

a Find the distance AC, to the nearest metre.

b Find the distance BC, to the nearest metre.

c Hence, find the distance covered by the jet in that one minute, to the nearest metre.

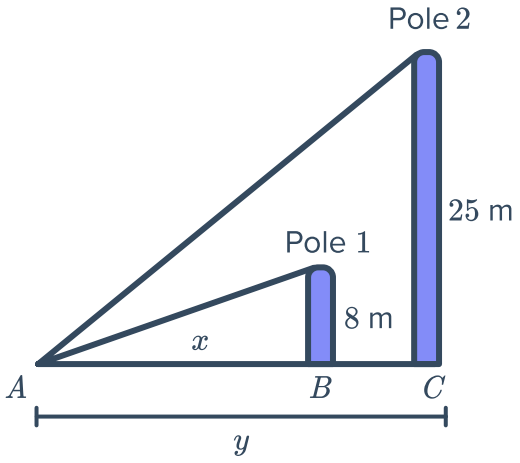


8 A man stands at point A looking at the top of two poles. Pole 1 has a height $8\ \text{m}$ and an angle of elevation of 34° from point A. Pole 2 has a height $25\ \text{m}$ and an angle of elevation of 57° from point A.

a Find x, the distance from A to B, to two decimal places.

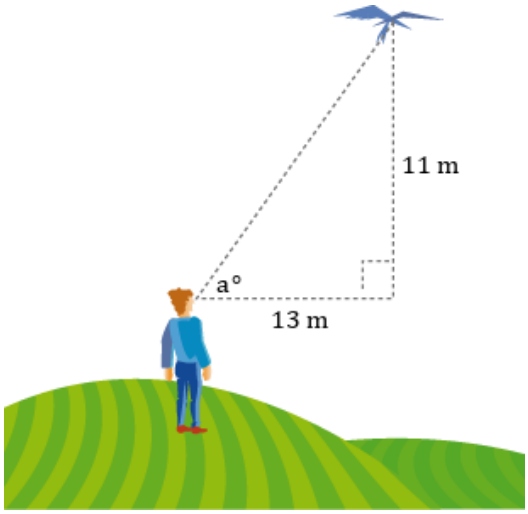
b Find y, the distance from A to C, to two decimal places.

c Hence, find BC, the distance between the two poles in meters. Round your answer to one decimal place.

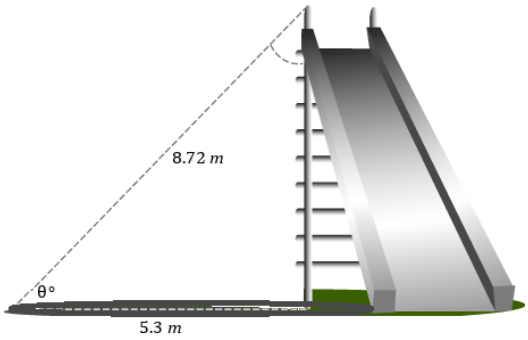


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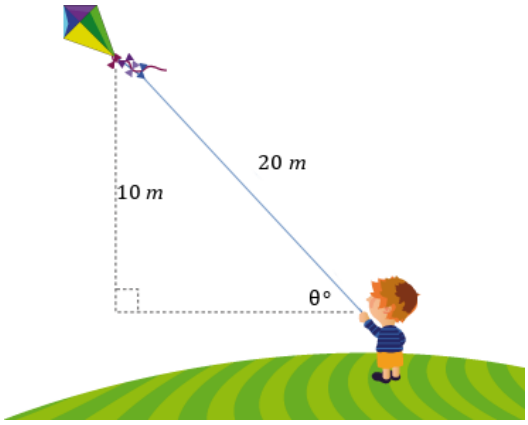
9 The person in the picture sights a paraglider above him. Find a to two decimal places.



10 A slide casts a shadow $5.3\ \text{m}$ along the ground. The distance between the tip of the shadow and the top of the slide is $8.72\ \text{m}$. Find θ to two decimal places.

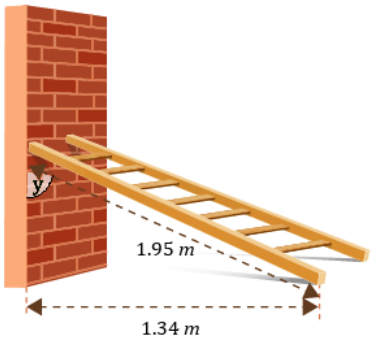


11 A boy flying his kite releases the entire length of his string which measures $20\ \text{m}$, so that the kite is $10\ \text{m}$ above him.
If the angle the string makes with the horizontal ground is θ, find θ to two decimal places.

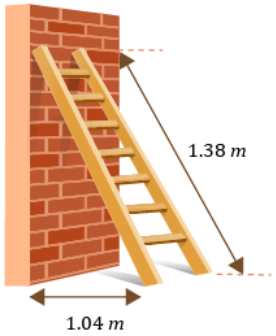


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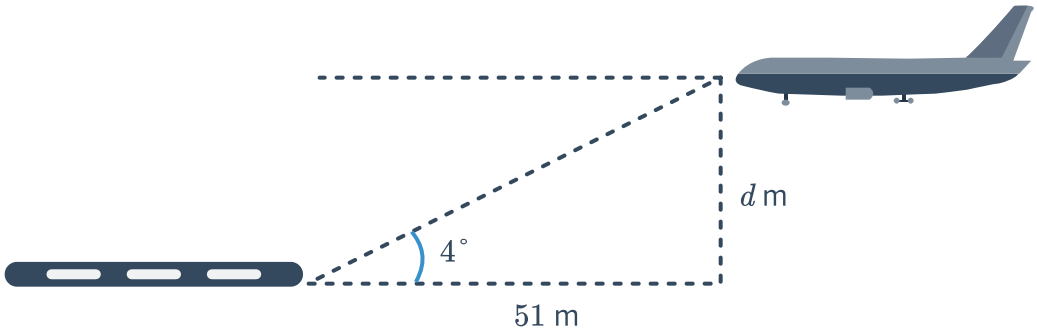
12 A ladder measuring $1.95\ \text{m}$ in length is leaning against a wall. If the angle the ladder makes with the wall is y°, find y to two decimal places.



13 A ladder measuring $1.38\ \text{m}$ in length is leaning against a wall. If the angle the ladder makes with the ground is y°, find y to two decimal places.



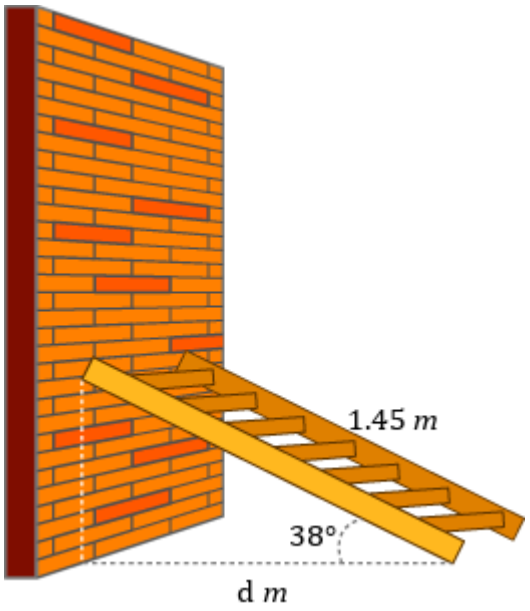
14 The final approach of an airplane when landing requires an angle of descent of about 4°. If the plane is directly above a point $51\ \text{m}$ from the start of the runway, find d, the height of the plane above the ground to the nearest meter.



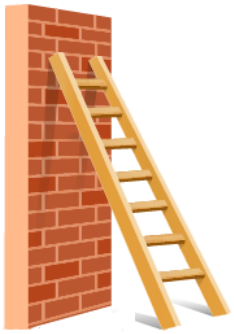
The final approach of an airplane when landing requires an angle of descent of about 4°.

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15 If d is the distance between the base of the wall and the base of the ladder, find the value of d to two decimal places.



16 A ladder of length $157\ \text{cm}$ needs to make an angle with the wall of between 10° and 20° for it to be safe to stand on.



a If the base of the ladder is $85\ \text{cm}$ from the wall, and the angle the ladder makes with the wall is x°, find x to two decimal places.

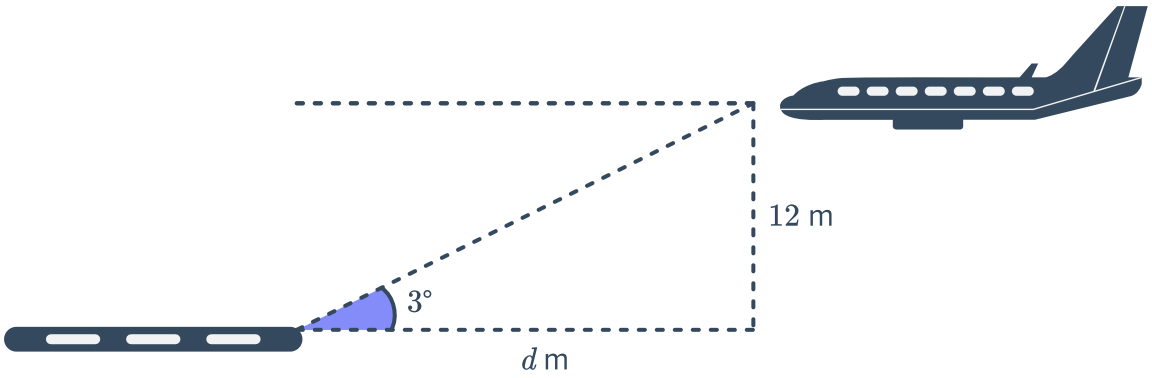
b If the base of the ladder is $17\ \text{cm}$ from the wall, and the angle the ladder makes with the wall is y°, find y to two decimal places.

c If the base of the ladder is $28\ \text{cm}$ from the wall, and the angle the ladder makes with the wall is z°, find z to two decimal places.

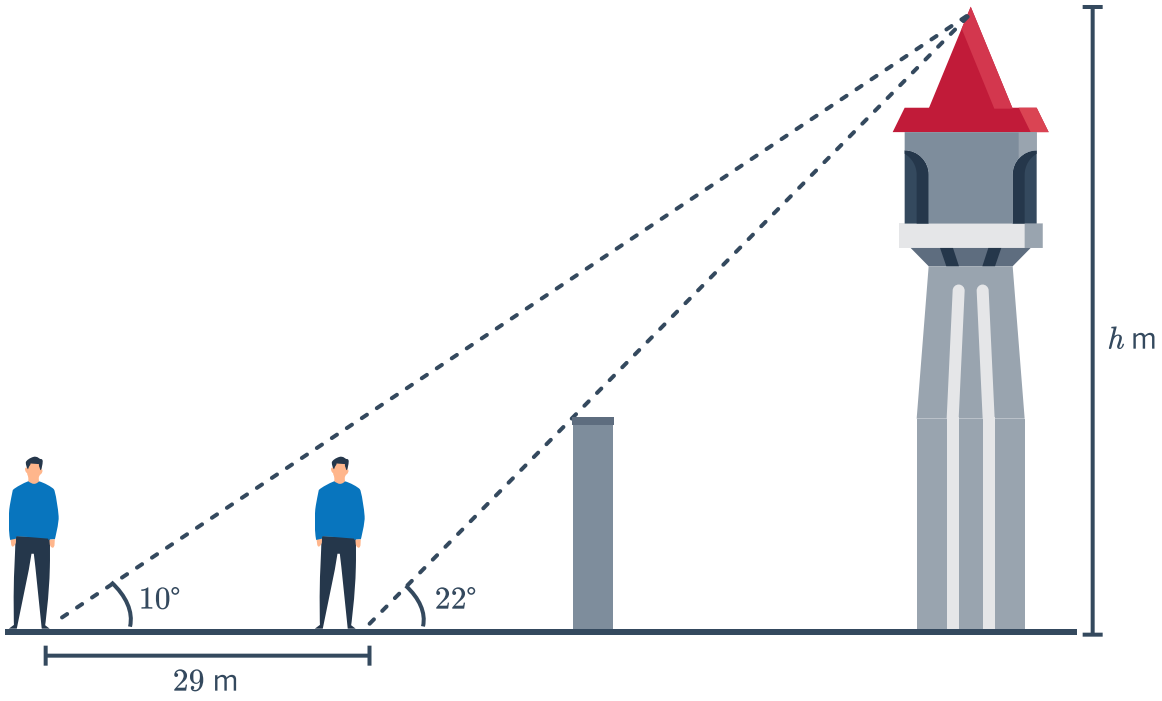
d At which length from the wall is the ladder safe?

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17 The final approach of an airplane when landing requires the pilot to adjust the angle of descent to about 3° as shown in the diagram below. If the plane is $12\ \text{m}$ above the runway and has $d\ \text{m}$ until touchdown, find d to the nearest meter.



18 A safety fence is constructed to protect tourists from the danger of an eroding castle toppling down. The surveyor takes an angle measurement to the top of the tower of 10°. She then walks $29\ \text{m}$ towards the tower and takes another reading of 22°.
Find the value of h to the nearest meter.



19 Amelia measures the angle of elevation to the top of a tree from a point, $29\ \text{m}$ away from the base, to be 31°. Find the height of the tree, h, to the nearest meter.

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20 The angle of elevation from an observer to the top of a tree is 18°. If the distance between the tree and the observer is d m and the tree is known to be 3.53 m high, find the value of d to two decimal places.

21 At a certain time of the day a light post, 6 m tall, has a shadow of 5.8 m. If the angle of elevation of the sun at that time is θ°, find θ to two decimal places.

22 From the top of a rocky ledge 188 m high, the angle of depression to a boat is 13°. If the boat is d m from the foot of the cliff, find the value of d correct to two decimal places.

23 Lisa is on a ship and observes a lighthouse on a cliff in the distance. The base of the cliff is 906 m away from the ship, and the angle of elevation of the top of the lighthouse from Lisa is 16°.

a If the top of the lighthouse is x m above sea level, find the value of x correct to two decimal places.

b If the lighthouse is 21 m tall, how tall is the cliff? Round your answer to two decimal places.

24 Buzz is standing 49 m from a building and measures the angle of elevation of the top of the building to be 23°.

a If the difference in height between the top of the building and Buzz's eye is h m , find the value of h correct to two decimal places.

b If Buzz's eye is 135 cm from the ground, what is the height of the building? Round your answer to one decimal place.

25 From the top of a cliff, 47 m above sea level, two boats are observed to be in the same direction. Their angles of depression from the top of the cliff are 12° and 23°.

a Find the distance between the base of the cliff and the boat furthest away, correct to two decimal places.

b Find the distance between the base of the cliff and the closest boat, correct to two decimal places.

c Hence, find the distance between the two boats in meters, correct to one decimal place.

26 A ship dropped anchor off the coast of a resort. The anchor fell 65 m to the sea bed. During the next 5 hours, the ship drifted 120 m. Calculate the angle of depression, x, between the anchor line and the surface of the water, rounded to the nearest degree.

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27 A $13.7\ \text{m}$ long string of lights joins the top of a tree to a point on the ground. If the tree is $3.7\ \text{m}$ tall, find θ, the angle the string of lights would make with the tree, rounded to two decimal places.

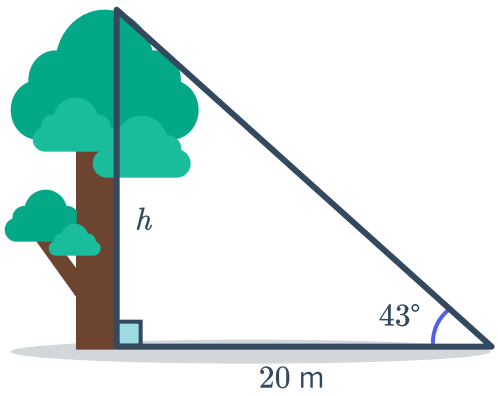
28 Jack is standing at the tip of a tree's shadow and knows that the angle from the ground to the top of the tree is 34°. If Jack is standing $29\ \text{m}$ away from the base of the tree, find the height of the tree to two decimal places.

29 The angle of elevation to the top of a $22\ \text{m}$ high statue is 54° from a point, A, due west of the statue. The point B is located $60\ \text{m}$ due south of point A.

- **a** Find the distance, x, from point A to the base of the statue, correct to two decimal places.
- **b** Hence, find y, the distance from point B to the base of the statue, correct to one decimal place.
- **c** Hence, find θ, the angle of elevation from point B to the top of the statue, correct to the nearest degree.

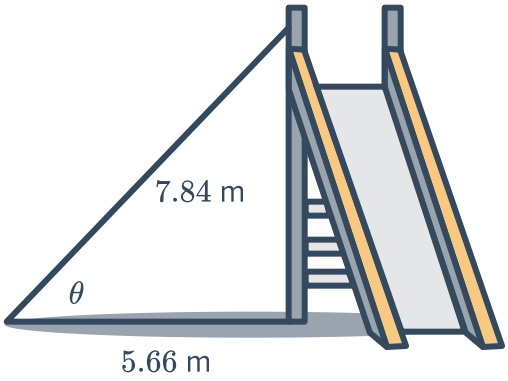
Additional questions

30 Aicha measures the angle of elevation to the top of a tree from a point $20\ \text{m}$ away to be 43°.
Find the height of the tree, h, to the nearest meter.

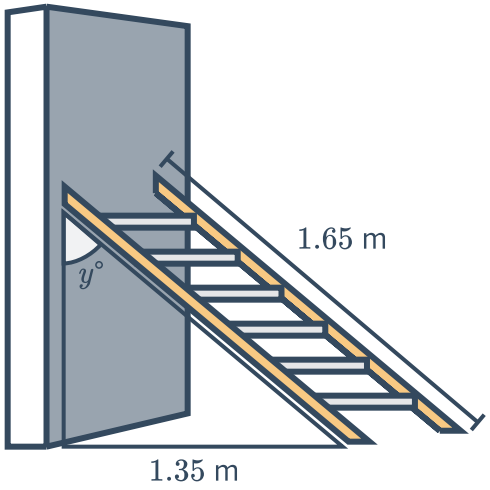


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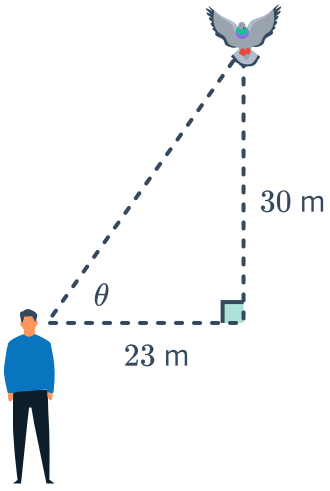
31 A slide casts a shadow $5.66\ \text{m}$ along the ground. The distance between the tip of the shadow and the top of the slide is $7.84\ \text{m}$. Find θ, rounding to two decimal places.



32 A ladder measuring $1.65\ \text{m}$ in length is leaning against a wall. If the angle the ladder makes with the wall is y°, find y, rounding to two decimal places.

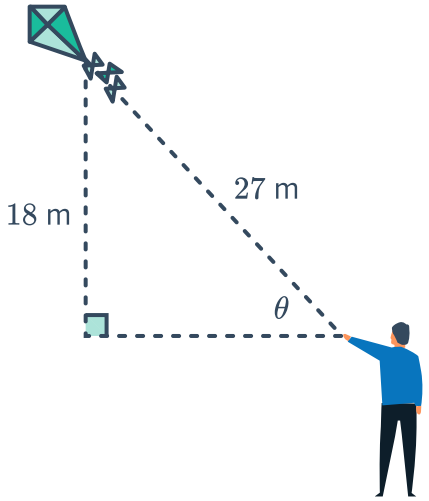


33 Xander sights a pigeon above him. Find θ, rounding to two decimal places.

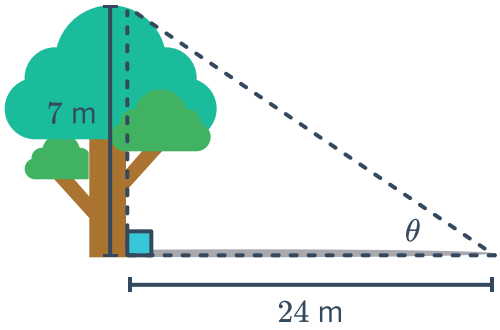


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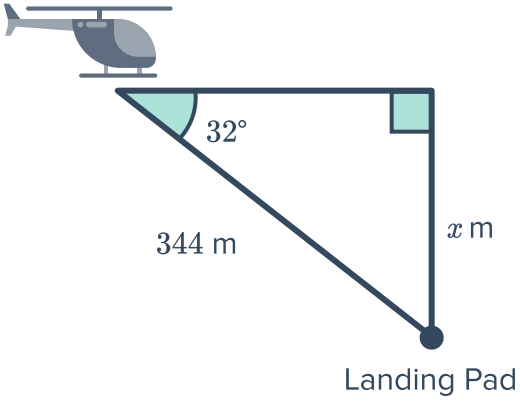
34 Saul is flying his kite and releases the entire length of his string which measures $27\ \text{m}$, so that the kite is $18\ \text{m}$ above him.
If the angle the string makes with the horizontal ground is θ, find θ, rounding to two decimal places.



35 During a particular time of the day, a tree casts a shadow of length $24\ \text{m}$. The height of the tree is estimated to be $7\ \text{m}$. Find the angle θ, formed by the length of the shadow and the arm extending from the edge of the shadow to the top of the tree. Round your answer to two decimal places.

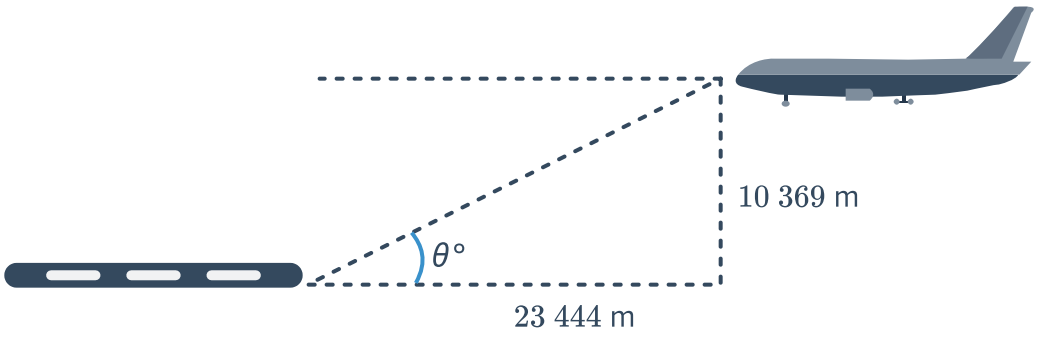


36 A helicopter is $344\ \text{m}$ away from its landing pad.
If the angle of depression to the landing pad is 32°, find x, the height of the helicopter above the ground, to the nearest meter.



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37 The airtraffic controller is communicating with a plane in flight approaching an airport for landing. The plane is $10\,369$ m above the ground and is still $23\,444$ m from the runway.
If θ° is the angle at which the plane should approach, find θ, rounding to the nearest tenth.



38 Glenda is riding her pushbike up a hill that has an incline of 5°.
If she rides her bike 817 ft up the hill, find d, the horizontal distance from where Glenda started, rounding to the nearest foot.

39 A ship dropped anchor off the coast of a resort. The anchor fell 83 ft to the sea bed. During the next 4 hours, the ship drifted 105 ft.
Find x, the angle between the anchor line and the surface of the water, rounding to the nearest degree.

40 A 27.3 m long cable joins the top of a flagpole to a point on the ground.
If the flagpole is 7.4 m tall, find θ, the angle the cable would make with the flagpole, rounding to two decimal places.

41 From the cockpit of an airplane flying at an altitude of 3000 m, the angle of depression to the airport is 57°. The airplane continues to fly in the same straight line, and after a few minutes the angle of depression to the airport becomes 67°.

a Sketch a diagram of the situation.

b The horizontal distance between the cockpit at the first sighting and the point directly above the airport is x m.
Find the value of x, rounding your answer to the nearest tenth.

c The horizontal distance between the cockpit at the second sighting and the point directly above the airport is y m.
Find the value of y, rounding your answer to the nearest tenth.

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42 Valentina and Mateo are on opposite sides of a valley. They both look down into the valley between them and see a large black bear. Valentina is $75\ \text{ft}$ above the valley and sees the bear at an angle of depression of 49°. Mateo is $59\ \text{ft}$ higher than Valentina and the angle of elevation from the bear to Mateo is 40°.

a Assuming Valentina and Mateo are both standing at the edges of cliffs at each side of the valley, determine the width of the valley to the nearest foot.

b Calculate the distance between Valentina and Mateo. Round your answer to the nearest tenth

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